Arithmetic sequences

1 Match the keyword to its definition. Write the correct letter in each box

a	arithmetic sequence
b	common difference
c	initial term
d	successive terms

a	a sequence that increases or decreases by the same step
b	the constant additive rule from one term to the next
d	terms that are next to each other in a sequence
c	the first term in a sequence

2 Match the arithmetic sequences with their descriptions. Write the correct letter in each box

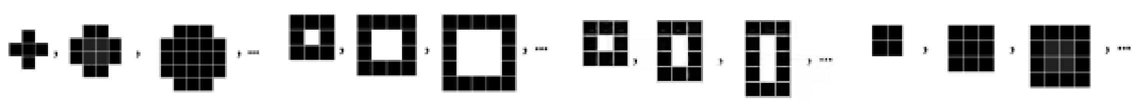
a	7, 5, 3, 1, …
b	7, 9, 11, 13, …
c	-1.6, -1.3, -1, -0.7, …
d	-1.6, -1.9, -2.2, -2.5, …
e	-2, 3, 8, 13, …
f	38, 33, 28, 23, …

a	initial term 7 and common difference -2
d	initial term -1.6 and common difference -0.3
c	initial term -1.6 and common difference 0.3
b	initial term 7 and common difference 2
f	ninth term -2 and common difference -5
e	fifth term 18 and common difference 5

3 Which of these sequences are arithmetic? Tick **2** correct answers

- [✓] $7, 2, -3, -8, \ldots$
- [ ] $4, 1, -1, -4, \ldots$
- [ ] $17, 24, 32, 39, \ldots$
- [✓] $\frac{4}{5}, \frac{1}{2}, \frac{1}{5}, -\frac{1}{10}, \ldots$
- [ ] $-1, -2, -4, -8, \ldots$

4 In which of these sequences do the number of shaded squares in each term form a linear sequence? Tick **2** correct answers



[ ] [✓] [✓] [ ]

5 An arithmetic sequence has 3rd term = 41 and 7th term = 65. What is the initial term? Fill in the blank

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6 Which of these sequences are arithmetic? Tick **4** correct answers

- [✓] $a, 2a, 3a, 4a, \ldots$
- [✓] $a+5, a+8, a+11, a+14, \ldots$
- [✓] $a-2b, a+b, a+4b, a+7b, \ldots$
- [✓] $5a+3, 3a+4, a+5, 6-a, \ldots$
- [ ] $a, 2a+4, 3a+6, 4a+8, \ldots$